

RetailIQ

**Web-Based Business Intelligence and
Predictive Analytics System**

Developer

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Year

2026

Project Type

Web Application (Django + Machine
Learning)

1. PROJECT OVERVIEW

Introduction

RetailIQ is a retail analytics web application designed to help businesses transform raw sales data into meaningful insights. The system enables users to upload retail datasets, analyze sales performance, monitor customer behavior, evaluate product trends, and generate business reports through an interactive dashboard. The application combines data analytics, machine learning, and visualization techniques to support data-driven decision-making.

Problem Statement

Retail businesses generate large volumes of transactional data every day. Analyzing this data manually is time-consuming and often fails to provide actionable insights. RetailIQ addresses this challenge by automating data processing, analysis, forecasting, and reporting, enabling businesses to make informed decisions quickly and efficiently.

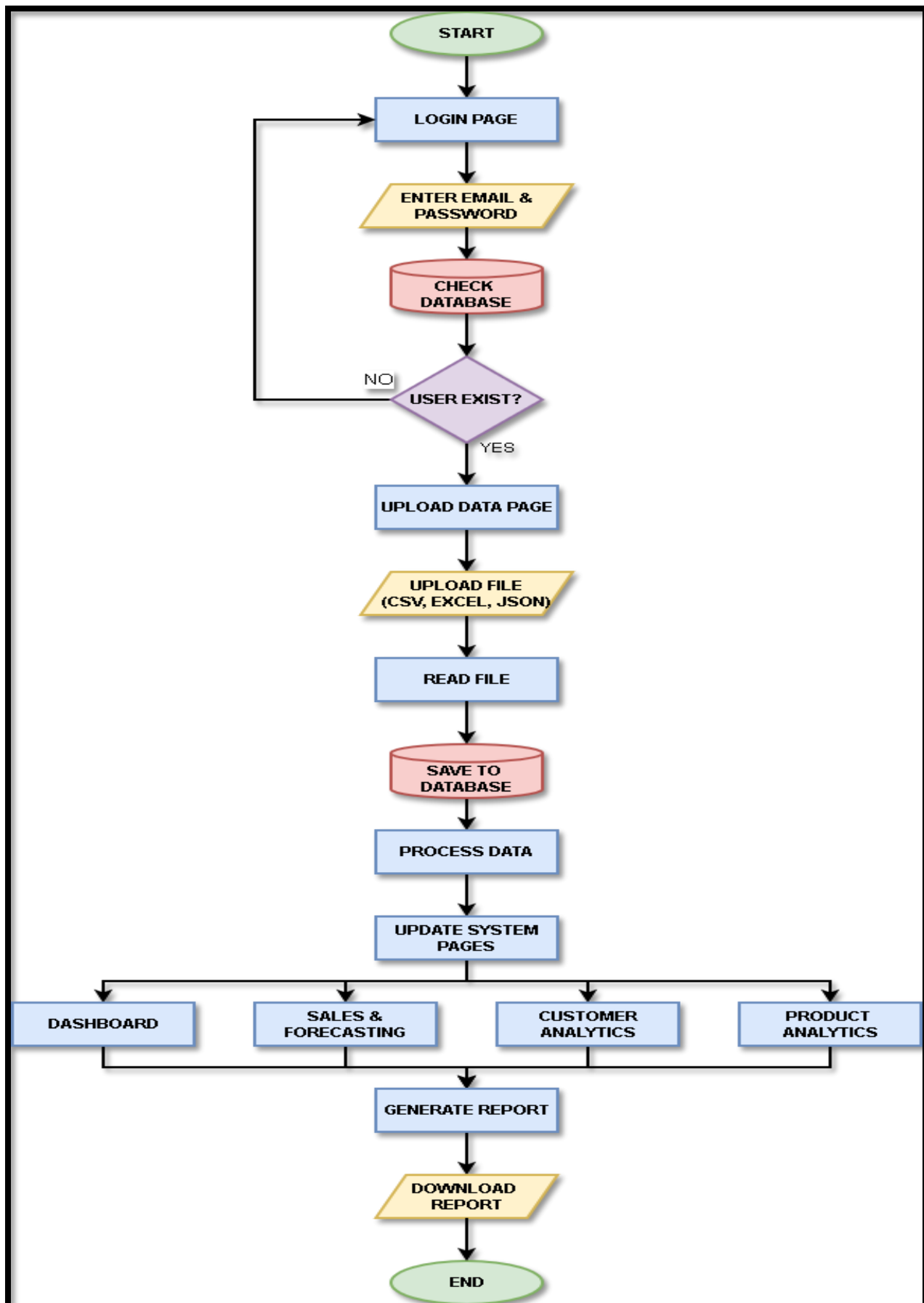
Objectives

- Provide Automate retail data processing and management.
- real-time business insights through dashboards.
- Analyze customer purchasing behavior.
- Identify top-performing and low-performing products.
- Forecast future sales trends using machine learning.
- Generate comprehensive business reports.

2. TECHNOLOGY STACK

Category	Technologies
Backend	Python, Django
Frontend	HTML, CSS, JavaScript, Bootstrap
Database	MySQL, PyMySQL
Data Processing	Pandas, NumPy
Machine Learning	Scikit-learn, Statsmodels
Reporting	ReportLab
Development Tools	Visual Studio Code

3. PROJECT FLOWCHART



4. DATABASE DESIGN (ER DIAGRAM)

Main Entities

Customers

- customer_id
- name
- email
- location
- signup_date

Products

- product_id
- product_name
- category
- selling_price
- stock_quantity

Orders

- order_id
- customer_id
- order_date
- total_amount

Order Items

- order_item_id
- order_id
- product_id
- quantity
- subtotal

Payments

- payment_id
- order_id
- payment_type
- payment_value

Marketing

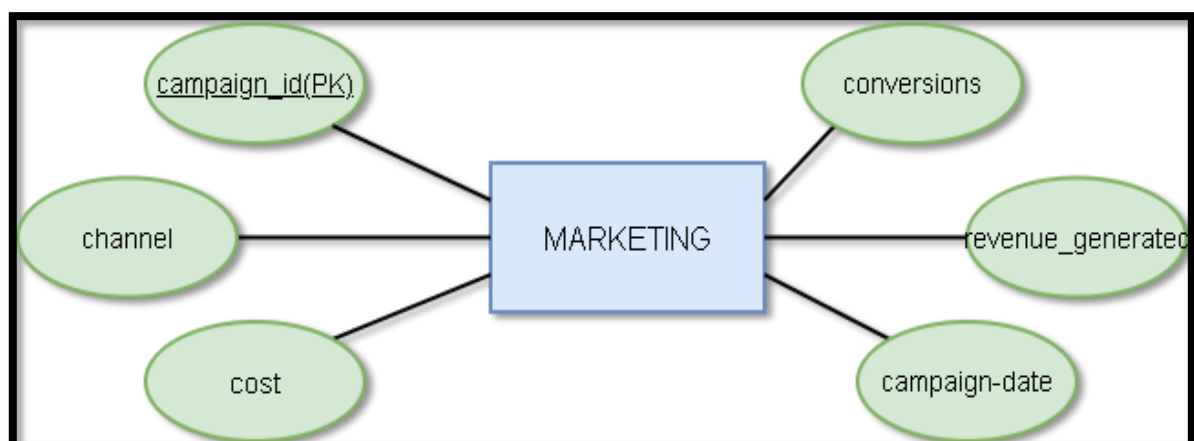
- campaign_id
- channel
- cost
- conversions
- revenue_generated

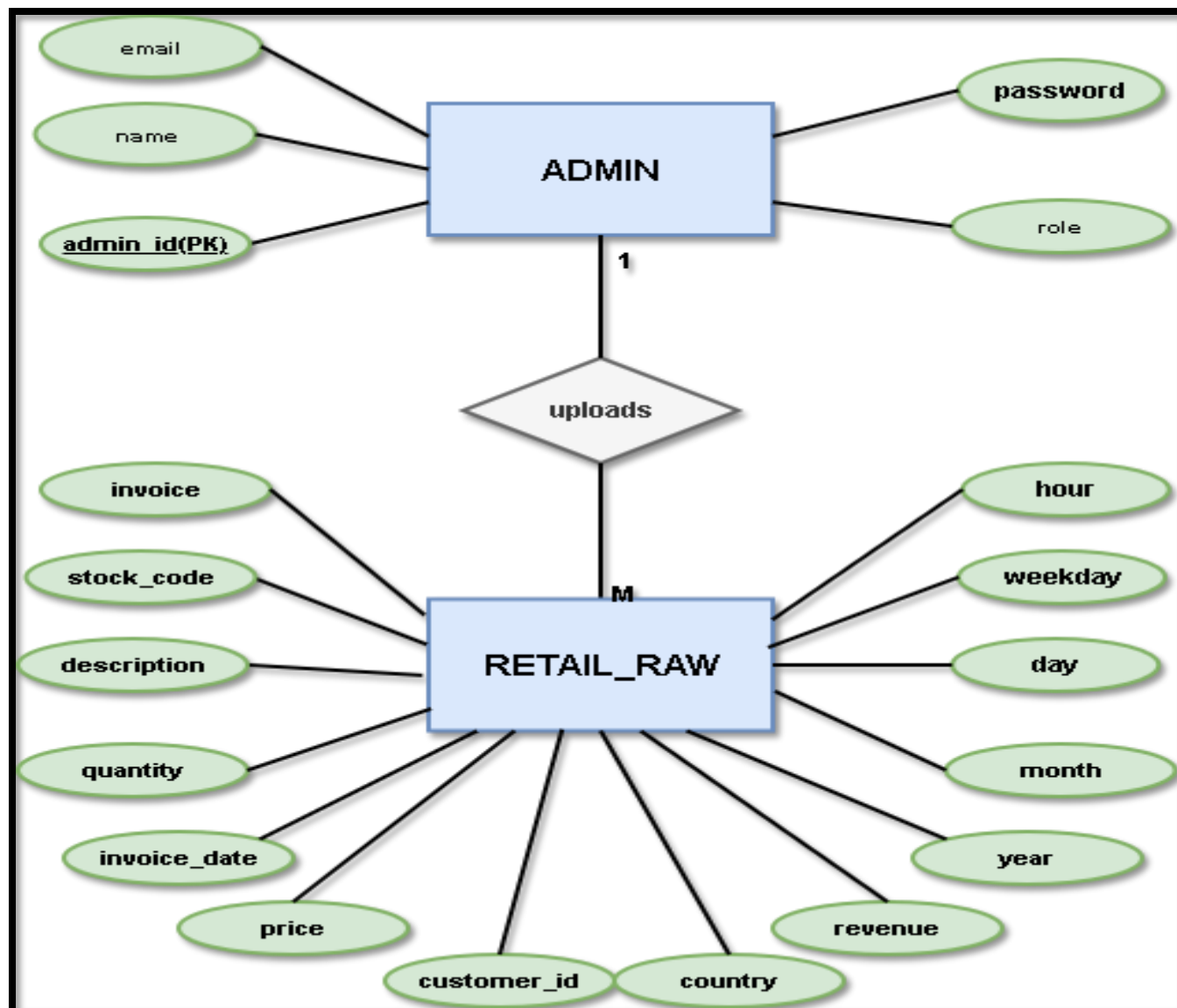
RetailRaw

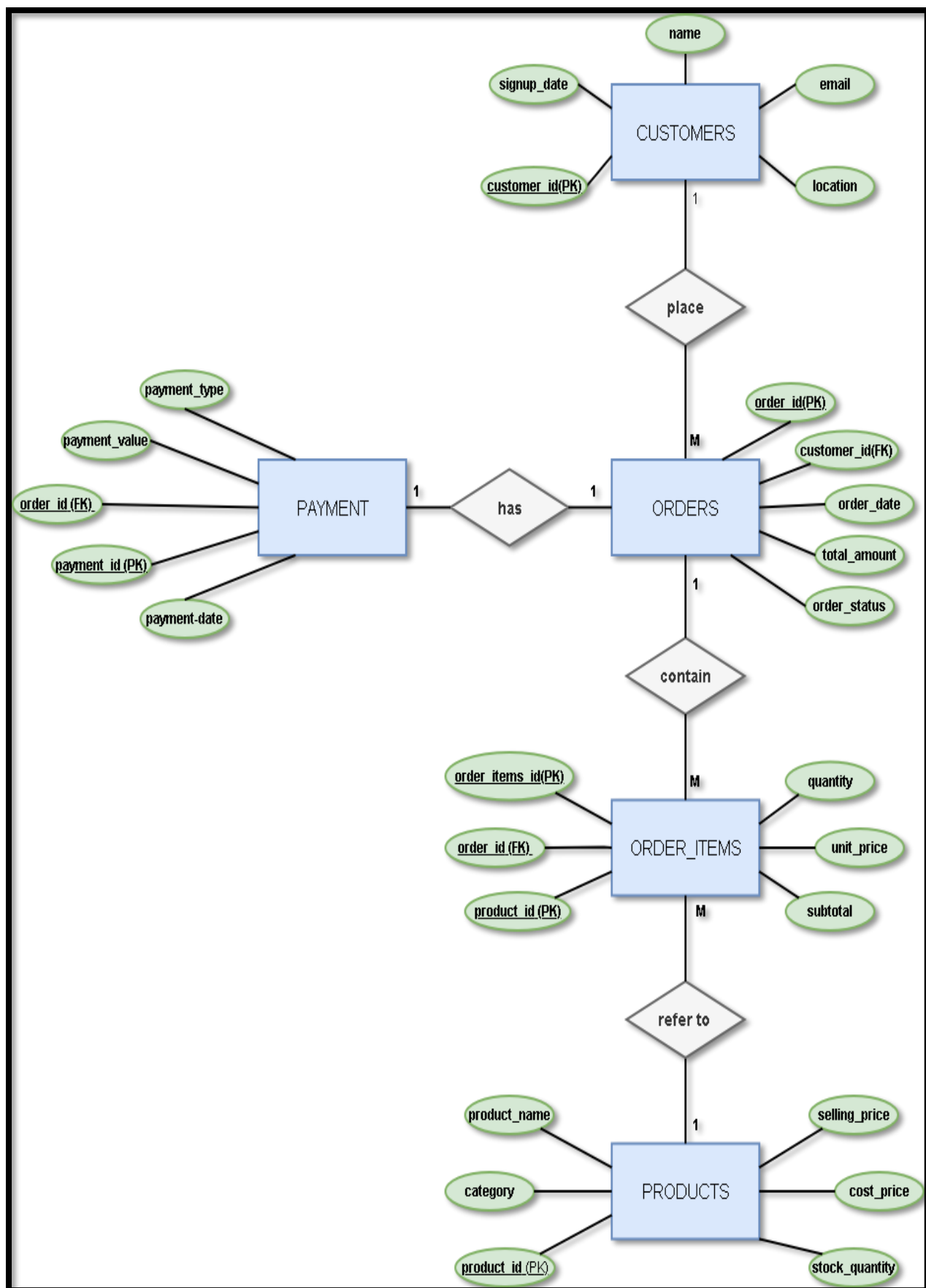
- invoice
- stock_code
- quantity
- price
- revenue

Relationships

- One Customer → Many Orders
- One Order → Many Order Items
- One Product → Many Order Items
- One Order → One Payment
- One Admin → Many Dataset Uploads







5. FEATURES

Authentication Module

- Secure admin login
- Session-based authentication

Data Upload Module

- Upload CSV files
- Upload Excel files
- Upload JSON files
- Automatic validation and processing

Dashboard Analytics

- Total Revenue
- Total Profit
- Total Orders
- Total Customers
- Monthly Sales Analysis
- Top Products Analysis

Sales Forecasting

- Sales trend analysis
- ARIMA-based forecasting
- Category-wise prediction
- Regional sales analysis

Customer Analytics

- Customer segmentation
- Customer lifetime value (CLV)
- Churn prediction
- New vs Returning customers

Product Analytics

- Inventory monitoring
- Low stock detection
- Best-selling products
- Worst-performing products
- Profit margin analysis
- Seasonal product trends

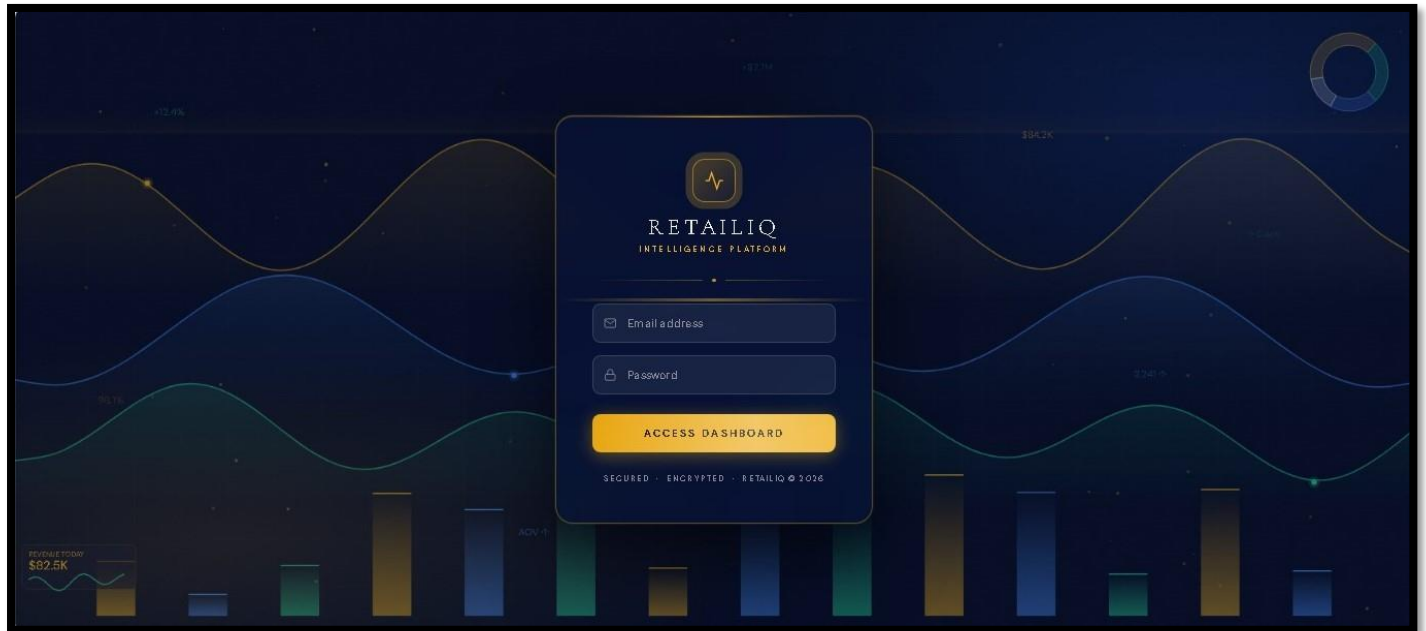
Report Generation

- Automated PDF reports
- KPI summaries
- Business recommendations

6. SCREENSHOTS

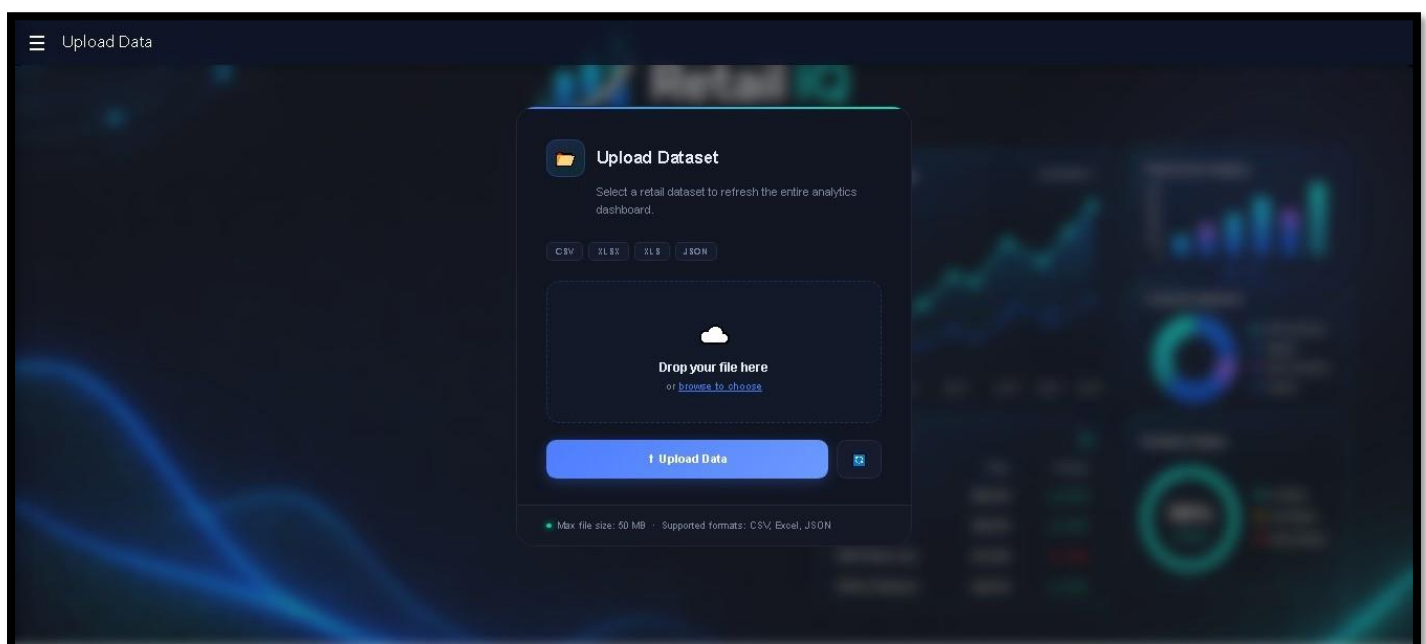
Login Page

Secure authentication system for administrators.



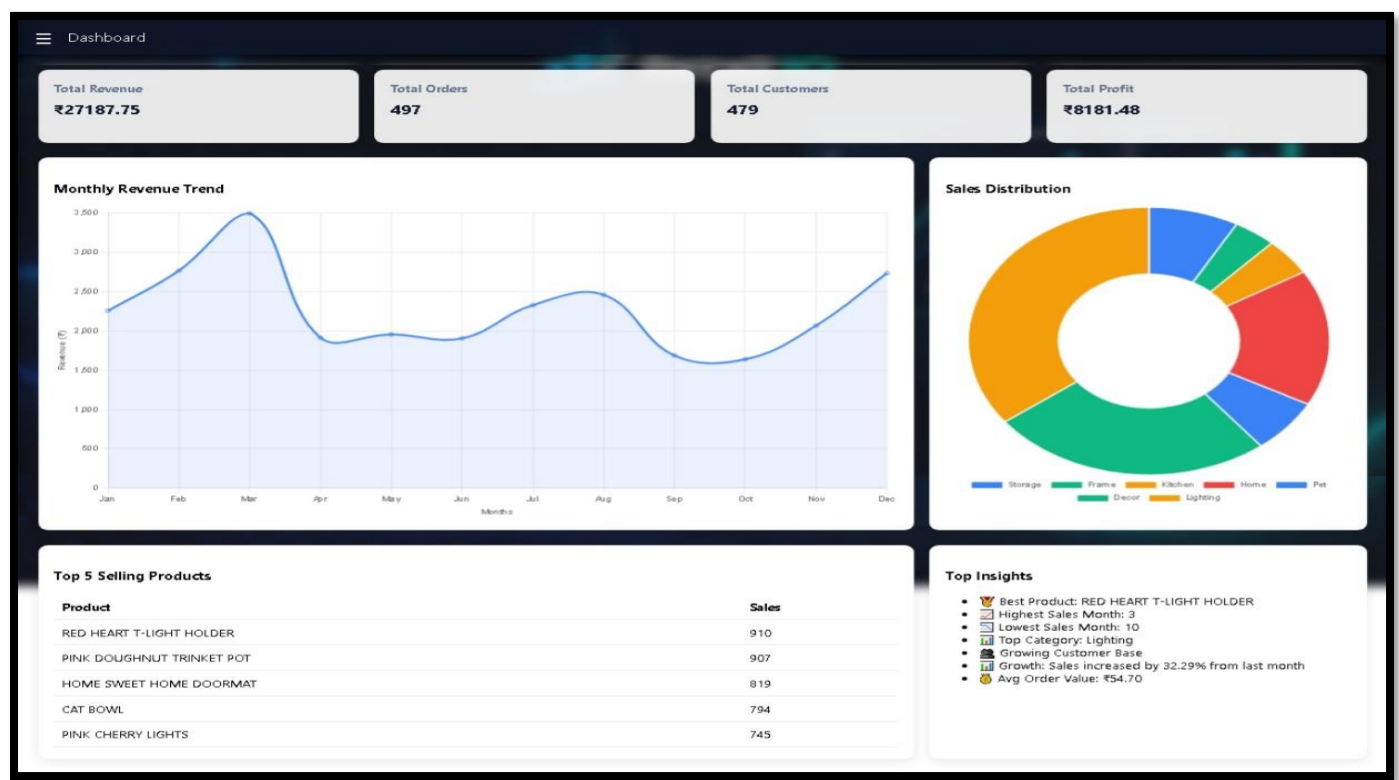
Upload Data Page

Allows uploading retail datasets in CSV, Excel, and JSON formats.



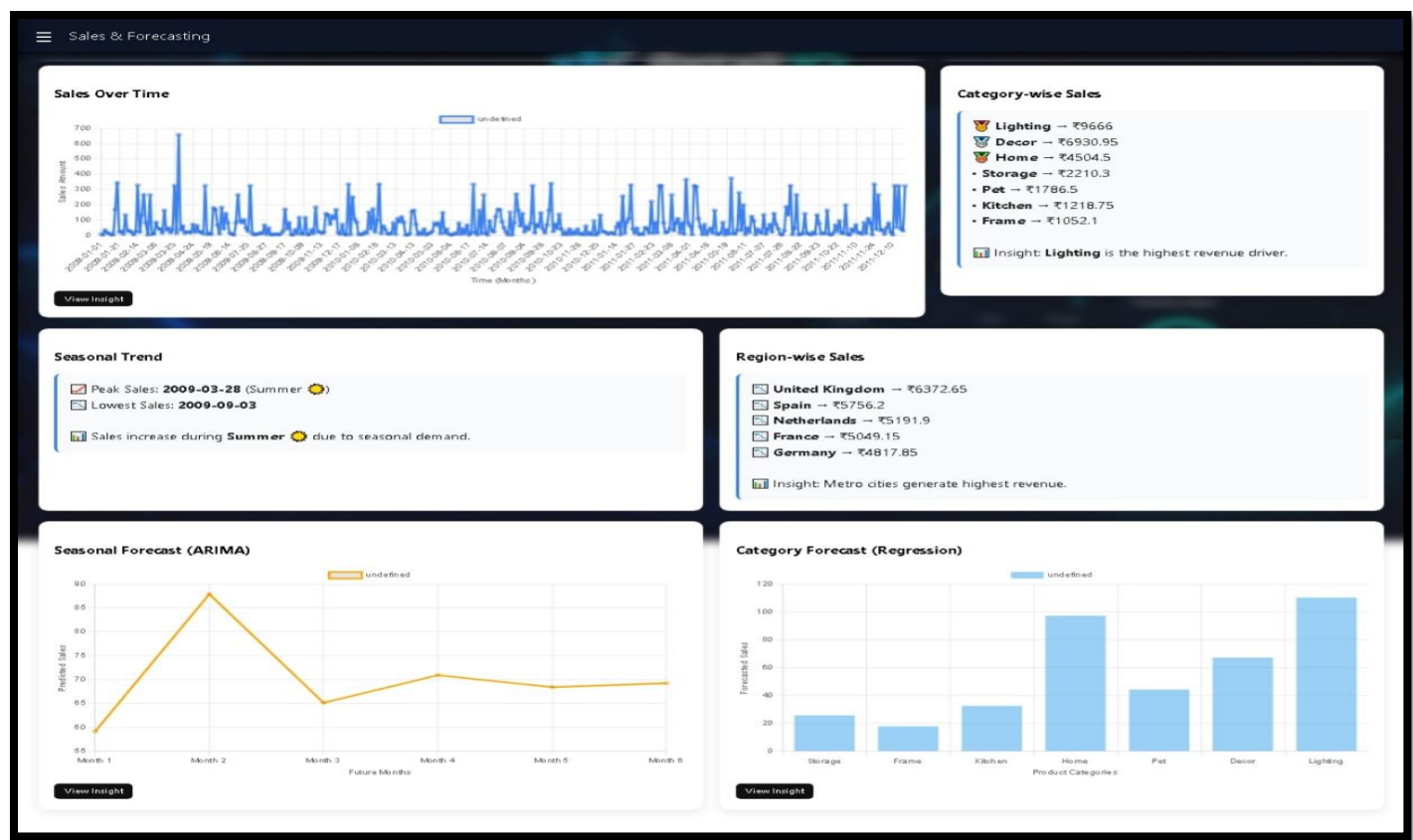
Dashboard Page

Displays key performance indicators and business metrics.



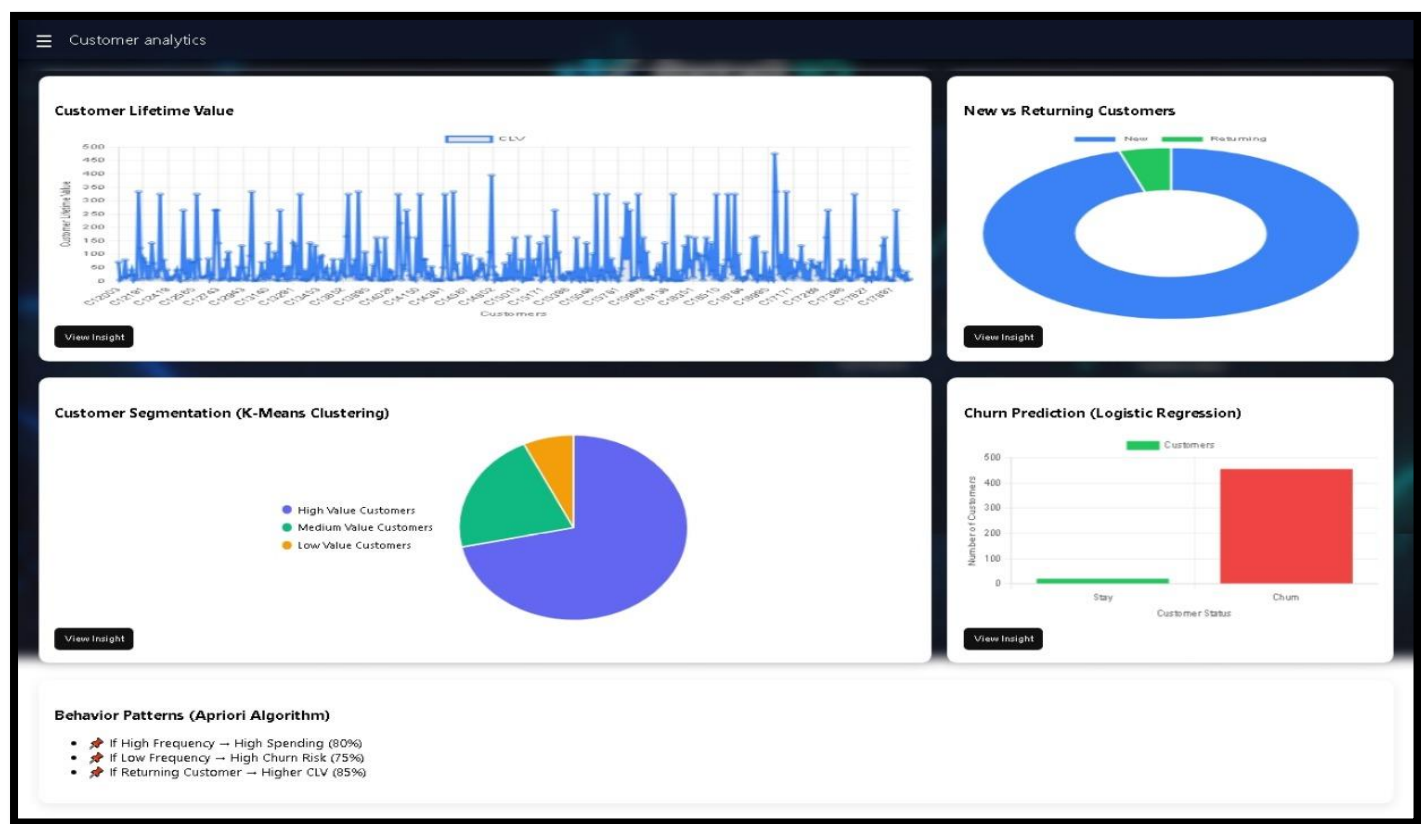
Sales & Forecasting Page

Shows sales trends, category performance, and future sales predictions.



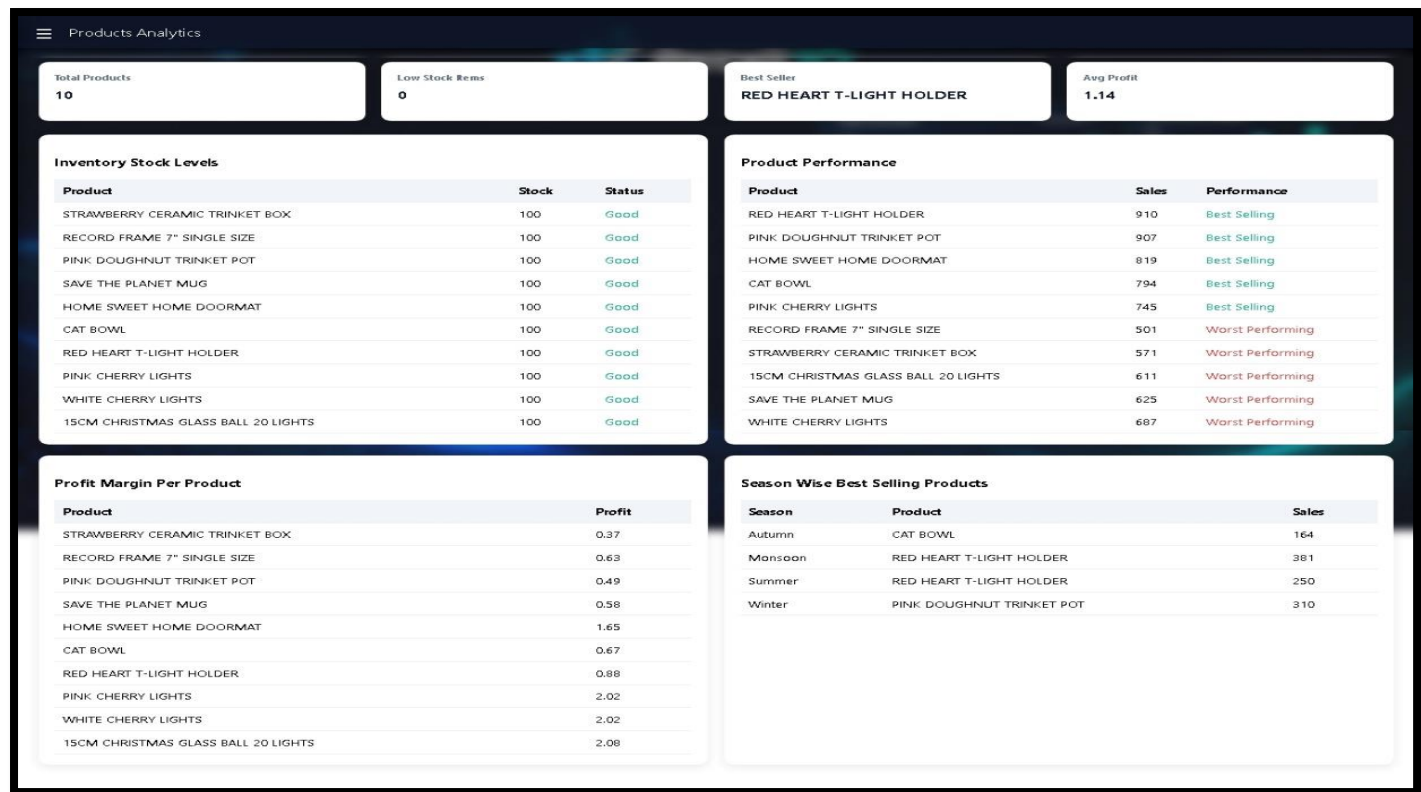
Customer Analytics Page

Provides segmentation, churn analysis, and customer lifetime value insights.



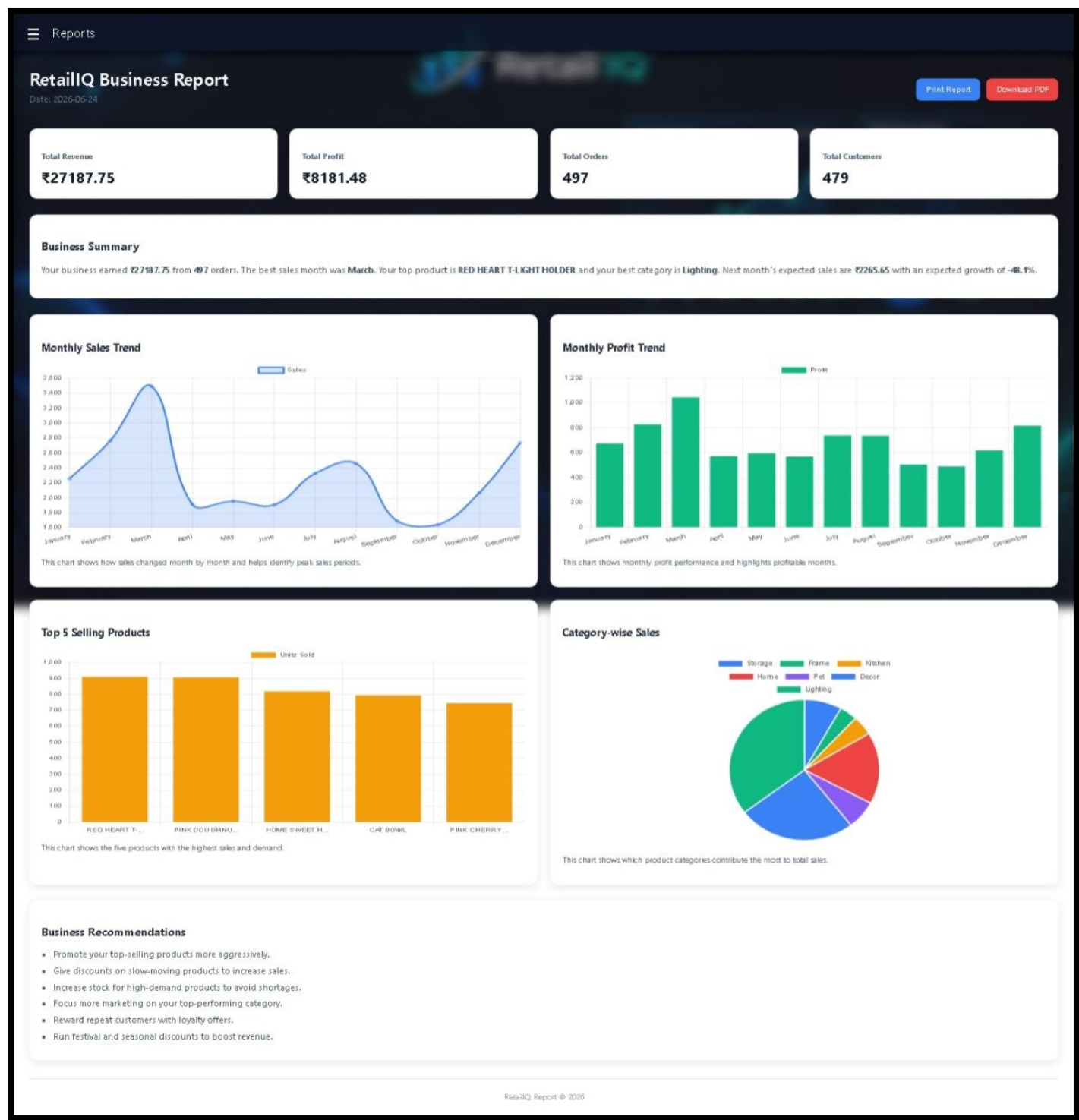
Product Analytics Page

Displays inventory status, best-selling products, and seasonal trends.



Report Page

Generates business reports and recommendations.



7. CHALLENGES FACED & SOLUTIONS

Challenge	Solution
Processing large retail datasets	Implemented Pandas-based data cleaning and transformation
Sales prediction accuracy	Applied ARIMA and Linear Regression models
Customer segmentation	Used K-Means clustering
Churn prediction	Implemented Logistic Regression
Dynamic report generation	Automated PDF generation using ReportLab

8. CONCLUSION

RetailIQ provides a complete solution for retail data management, analytics, forecasting, and reporting. By integrating Python, Django, MySQL, machine learning, and data visualization techniques, the system helps businesses gain actionable insights from their data. Features such as sales forecasting, customer segmentation, churn prediction, and automated reporting enable organizations to improve operational efficiency and make informed business decisions

9. GITHUB REPOSITORY

<https://github.com/PriyaYadav274/RetailIQ>